PREDICTIVE ANALYTICS SYSTEM FOR ANALYSIS OF SURAT BUSINESSES

Internship Project Report — Week 4

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1. Introduction

In today's digital economy, having an online presence has become essential for businesses of all sizes. Customers increasingly rely on the internet to search for products, compare services, read reviews, and locate nearby businesses. Despite this growing trend, many local businesses continue to operate without a dedicated website, limiting their visibility and opportunities for growth.

This project aims to address this challenge by developing a Predictive Analytics System that analyzes local business data and predicts which businesses are most likely to benefit from having a website. The project combines data analysis, business intelligence, and machine learning to transform raw business information into meaningful insights and predictive recommendations.

The analysis is based on Google Maps business data collected from various commercial areas of Surat. After cleaning and preprocessing the data, interactive dashboards and predictive models were developed to support data-driven decision-making. The project demonstrates how machine learning can assist in identifying digital transformation opportunities for local businesses.

2. Objectives

The primary objectives of this project are:

- To collect and analyze business data from Google Maps.
- To clean and preprocess the dataset for accurate analysis.
- To perform Exploratory Data Analysis (EDA) and identify business trends.
- To develop interactive Power BI dashboards for business visualization.
- To build and compare multiple machine learning models for prediction.
- To identify the best-performing predictive model based on evaluation metrics.
- To generate meaningful business insights and recommendations for improving digital presence.

3. Technology Stack

Technology	Purpose
Python	Data preprocessing, analysis, and machine learning
Jupyter Notebook	Data cleaning, EDA, and model development
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Matplotlib & Seaborn	Data visualization
Scikit-learn	Machine learning model development and evaluation
Power BI	Interactive dashboard creation

4. Dataset Overview

Dataset Description

The dataset used in this project was collected from Google Maps and contains information about businesses located in different commercial areas of Surat, Gujarat. The purpose of collecting this dataset was to analyze the digital presence of local businesses and identify those that could benefit from having a dedicated website.

The dataset includes information such as business name, area, address, category, customer rating, number of reviews, and website availability. These attributes were used for exploratory data analysis and machine learning model development.

Dataset Summary

Attribute	Value
Dataset Source	Google Maps
City	Surat, Gujarat
Total Records	691
Total Columns	10
Areas Covered	10
Business Categories	102
File Format	CSV

Dataset Attributes

Column Name	Description
Sr.No	Unique identifier for each business
Area	Business locality within Surat
Address/Street	Business address
Shop Name	Name of the business
Category	Type of business
Rating	Customer rating (out of 5)
No. of Reviews	Total customer reviews
Website (Yes/No)	Indicates whether the business has a website
Website Needed (Yes/No)	Target variable used for prediction
Opportunity Flag	Identifies highly rated businesses without a website

The Website Needed (Yes/No) column was used as the target variable for the machine learning models, while the remaining relevant attributes served as input features for prediction.

5. Data Cleaning and Preprocessing

Raw datasets often contain inconsistencies, missing values, and formatting issues that can affect the accuracy of analysis and machine learning models. Therefore, data cleaning and preprocessing were performed before visualization and model development.

The preprocessing was carried out using Python in Jupyter Notebook with the help of the Pandas library.

Data Loading

The dataset was imported into Jupyter Notebook using the Pandas library. After loading the data, the first few records, data types, and dataset dimensions were examined to understand the structure of the dataset.

Missing Value Analysis

The dataset was checked for missing values using the isnull() function.

Observations:

- Most columns were complete.
- Missing values, where present, were handled appropriately.
- Records with insufficient information for analysis were excluded from calculations when necessary.

This ensured that the dataset remained reliable for visualization and predictive modeling.

Duplicate Record Check

Duplicate business records can lead to misleading analytical results.

The dataset was examined using the duplicated() function, and duplicate entries were removed whenever necessary. This helped maintain the uniqueness of each business record.

Data Standardization

To maintain consistency across the dataset, several preprocessing operations were performed:

- Removed unnecessary white spaces.
- Standardized text formatting.
- Corrected inconsistent category names.
- Verified numerical values for ratings and review counts.
- Standardized Website status into Yes and No.

These steps improved the overall quality and consistency of the dataset.

Feature Engineering

Two additional columns were created to support business analysis and predictive modeling.

Website Needed (Yes/No)

This column identifies businesses that are likely to require a website based on predefined business conditions.

Opportunity Flag

This column highlights businesses with high customer ratings but no website, helping identify potential opportunities for digital transformation.

These newly created features improved both the business analysis and the machine learning process.

Data Encoding

Machine learning algorithms require numerical input. Therefore, categorical variables such as Area, Category, Website Status, and Address were converted into numerical values using Label Encoding.

This transformation allowed the machine learning models to process categorical information efficiently.

Train-Test Split

The prepared dataset was divided into training and testing datasets using the 80:20 ratio.

Dataset	Size
Training Records	552
Testing Records	139

The training dataset was used to build the prediction models, while the testing dataset was used to evaluate their performance on unseen data.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset and identify meaningful business patterns before developing predictive models.

Several visualizations were created using Matplotlib, Seaborn, and Power BI.

Website Availability Analysis

The analysis revealed that website adoption among local businesses is relatively low.

- Businesses with Website: 156
- Businesses without Website: 535

This indicates that 77.42% of businesses currently operate without a dedicated website.

Interpretation

The findings suggest a significant opportunity for businesses to improve their online visibility. Establishing a website can help attract new customers, improve brand credibility, and increase accessibility in the digital marketplace.

Business Category Analysis

The distribution of business categories shows that Clothing Stores, Medical Stores, Electronics Stores, and Grocery Stores are among the most common businesses in the dataset.

Interpretation

These categories dominate the local market and represent the largest potential audience for digital transformation initiatives.

Area-wise Business Distribution

The dataset covers businesses across ten different areas of Surat.

Areas such as Palanpur Canal Road, Rander Road, and Gujarat Gas Circle contain the highest concentration of businesses.

Interpretation

These commercial areas represent ideal locations for targeted digital marketing campaigns and website development initiatives.

Customer Rating Analysis

The average customer rating across all businesses is 4.35 out of 5, indicating a generally positive customer experience.

Interpretation

Although customer satisfaction is already high, many businesses still lack an online presence. This suggests that the main challenge is digital visibility rather than service quality.

Opportunity Analysis

The analysis identified 46 businesses with high customer ratings but no website.

Interpretation

These businesses represent high-potential candidates for website development because they already have strong customer trust and are likely to benefit from increased online exposure.

7. Power BI Dashboard Explanation

An interactive Power BI dashboard was developed to visualize business information and machine learning results. The dashboard enables users to explore business distribution, identify digital opportunities, and understand predictive model performance through interactive charts and KPIs.

The dashboard consists of four pages, each designed to provide different levels of business analysis and decision support.

Dashboard 1 — Business Overview

The Business Overview dashboard presents a summary of the entire dataset using Key Performance Indicators (KPIs) and visual charts.

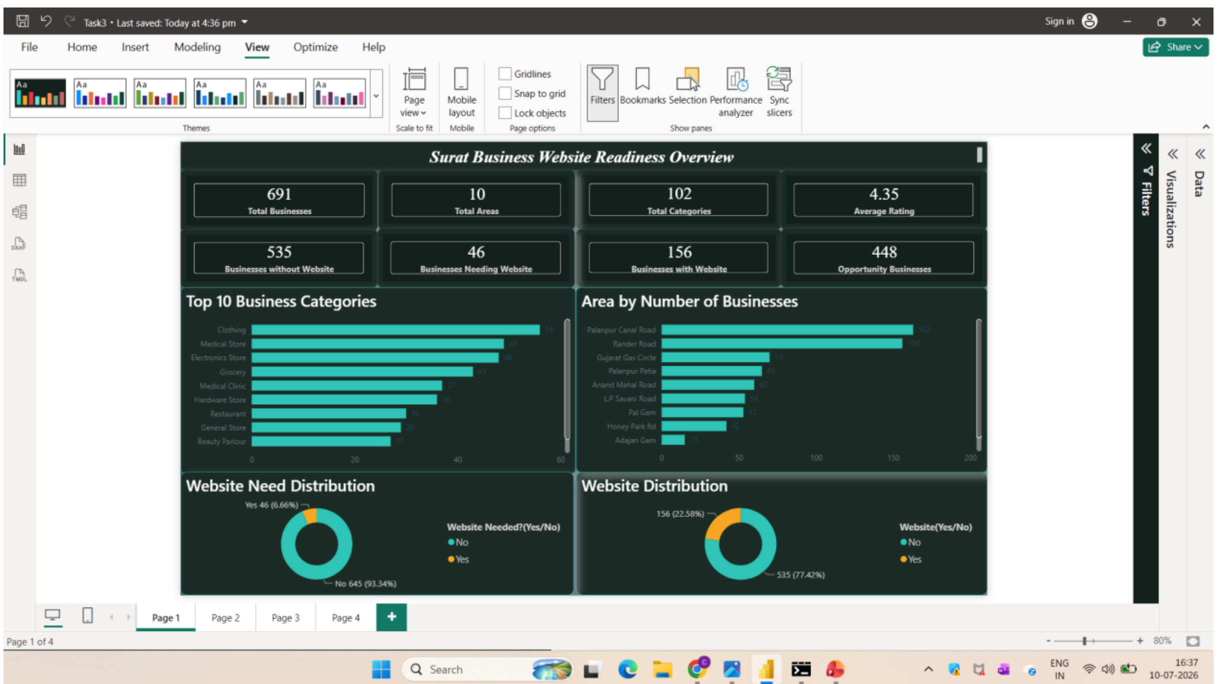

Figure 1 — Power BI Dashboard 1: Business Overview.

Dashboard 2 — Predictive Model Results

The second dashboard focuses on the machine learning models developed during the project.

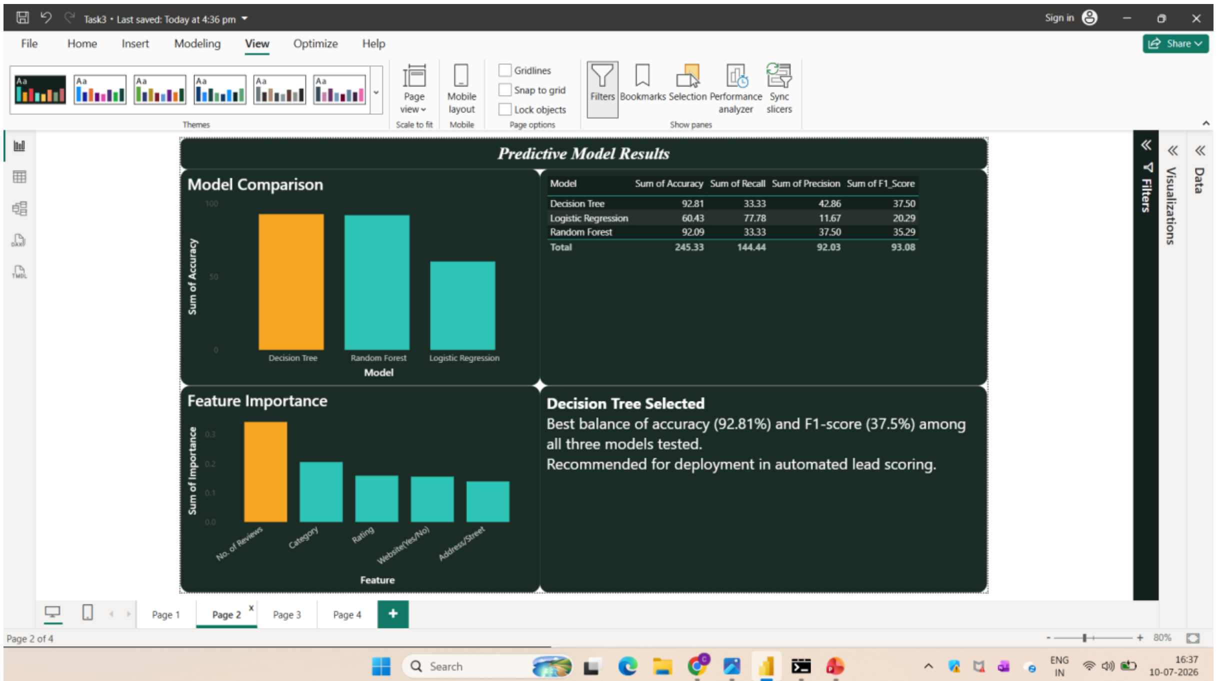

Figure 2 — Power BI Dashboard 2: Predictive Model Results.

Dashboard 3 — Business Opportunity Analysis

This dashboard identifies businesses with high potential for digital transformation.

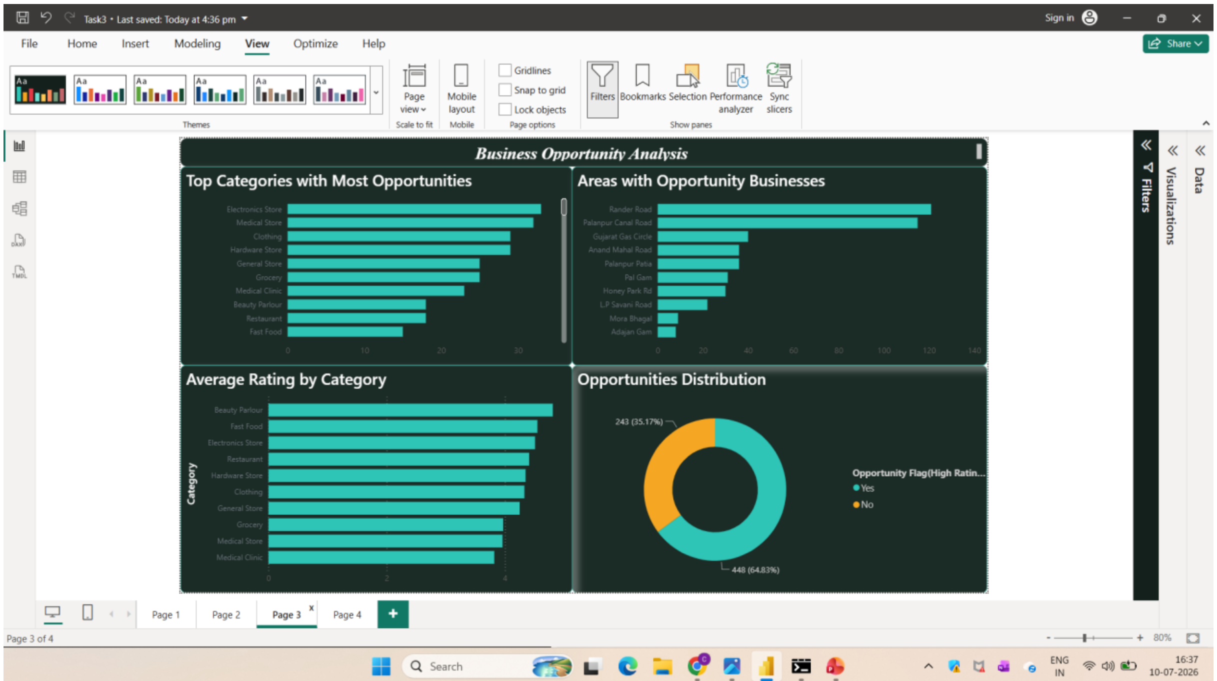

Figure 3 — Power BI Dashboard 3: Business Opportunity Analysis.

Dashboard 4 — Business Insights

The final dashboard presents key findings and recommendations derived from the analysis.

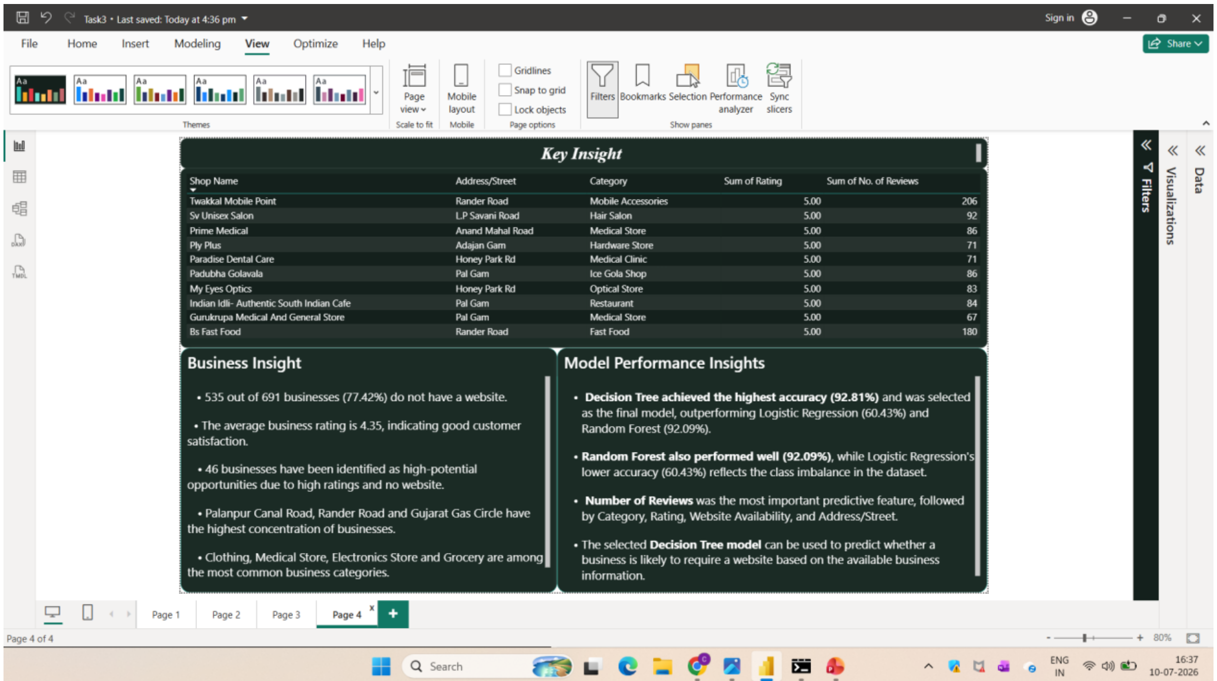

Figure 4 — Power BI Dashboard 4: Business Insights.

8. Machine Learning Implementation

Machine Learning was applied to predict whether a business is likely to require a website based on its characteristics. Predictive modeling enables organizations to identify businesses with high digital growth potential without relying solely on manual analysis.

Target Variable

The prediction target for this project is: Website Needed (Yes/No)

This binary variable indicates whether a business is considered to require a website based on predefined business conditions.

Input Features

The following features were selected for model training:

- Area
- Address/Street
- Business Category
- Customer Rating
- Number of Reviews
- Website Availability

These variables provide meaningful information about the business and influence the prediction of website necessity.

Data Preparation

Before training the machine learning models, several preprocessing steps were performed:

- Label Encoding of categorical variables
- Feature selection
- Train-Test Split (80:20)
- Data validation

The processed dataset contained:

Dataset	Records
Training Data	552
Testing Data	139

9. Machine Learning Models

Three supervised machine learning algorithms were implemented and evaluated.

Logistic Regression

Logistic Regression is a statistical classification algorithm commonly used for binary prediction problems.

In this project, Logistic Regression was trained to classify whether a business requires a website.

Result: Accuracy — 60.43%

Although Logistic Regression successfully classified many businesses, its overall prediction performance was lower than the tree-based models because the dataset contains complex relationships that cannot be represented well using a linear decision boundary.

Decision Tree

Decision Tree is a supervised learning algorithm that predicts outcomes by dividing the dataset into smaller subsets using decision rules.

The model learns patterns from business attributes and generates predictions through a tree-like structure.

Result: Accuracy — 92.81%

Among all evaluated models, the Decision Tree achieved the highest accuracy and demonstrated excellent prediction performance. It effectively captured the relationships between business characteristics and website requirements.

Therefore, the Decision Tree model was selected as the final predictive model for this project.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction stability.

Result: Accuracy — 92.09%

The Random Forest model also achieved high prediction accuracy. However, its performance was slightly lower than the Decision Tree model on this dataset.

10. Model Evaluation (Model Comparison)

The performance of all machine learning models was compared using classification accuracy.

Model	Accuracy
Logistic Regression	60.43%
Decision Tree	92.81%
Random Forest	92.09%

Analysis

The comparison clearly shows that the Decision Tree model outperformed the other models, achieving the highest prediction accuracy of 92.81%. Its ability to capture complex decision rules and nonlinear relationships made it the most suitable model for this dataset.

11. Feature Importance Analysis

Feature Importance helps identify which variables have the greatest influence on the prediction.

According to the Decision Tree model, the most important features are:

1. Number of Reviews
2. Business Category
3. Customer Rating
4. Website Availability
5. Address/Street

Explanation

Number of Reviews was identified as the most influential feature, indicating that businesses with greater customer engagement are more likely to require a website. Business Category also plays a significant role because different industries have varying levels of digital adoption. Customer Rating reflects business reputation, while Website Availability and Address/Street provide additional contextual information that improves prediction accuracy.

The Feature Importance analysis helps explain why the model makes certain predictions and provides valuable insights for businesses planning digital transformation initiatives.

12. Business Insights

Insight 1: Most Businesses Still Don't Have a Website

Out of **691 businesses**, **535 businesses (77.42%)** do not have their own website. Even though many of them have good customer ratings, they still depend mainly on offline customers or Google Maps for their online visibility. This shows a huge opportunity for businesses to improve their digital presence. A website can help them reach more customers, increase trust, and improve business growth.

Insight 2: Decision Tree Performed the Best

Three machine learning models were developed and compared. The **Decision Tree model achieved the highest accuracy of 92.81%**, making it the best-performing model, followed by Random Forest (92.09%) and Logistic Regression (60.43%). The Decision Tree model was able to identify website requirements more accurately than the other models. It can be used as a reliable tool for predicting which businesses are likely to benefit from having a website.

Insight 3: Customer Reviews Play an Important Role

Feature Importance analysis showed that the **Number of Reviews** had the greatest impact on the prediction results. Businesses with more customer reviews usually have higher customer engagement. These businesses are more likely to gain additional benefits from having a professional website.

Insight 4: High-Rated Businesses Have Better Growth Potential

The average business rating in the dataset is **4.35**, which shows that most businesses provide good customer service. However, many of these highly rated businesses still do not have a website.This suggests that businesses with a good reputation should invest in building an online presence to attract more customers and expand their business.

Insight 5: Business Category and Area Influence Website Readiness

Categories such as **Clothing Stores, Medical Stores, Electronics Stores, and Grocery Stores**, along with areas like **Palanpur Canal Road, Rander Road, and Gujarat Gas Circle**, have the highest number of businesses and website opportunities.

These categories and locations should be given priority for website development and digital marketing initiatives, as they have strong potential for business growth.

13. Business Recommendations

- **Encourage businesses without websites** to build an online presence to improve their visibility, attract more customers, and strengthen their digital identity.

- **Give priority to highly rated businesses with strong customer reviews**, as they have a higher potential to benefit from website development and online marketing.

- **Use the Decision Tree model** for predicting website requirements, as it achieved the highest accuracy (**92.81%**) and proved to be the most reliable model in this project.

- **Focus website development and digital awareness programs** in business-dense areas such as **Palanpur Canal Road, Rander Road, and Gujarat Gas Circle**, where the opportunities for digital transformation are higher.

- **Develop category-specific website solutions** for businesses such as Clothing Stores, Medical Stores, Electronics Stores, and Grocery Stores to better meet their customer needs and improve online accessibility.

- **Regularly update and retrain the machine learning model** using new business data to maintain prediction accuracy and adapt to changing business trends.

15. Conclusion

This project successfully developed a Predictive Analytics System for analyzing the digital presence of local businesses in Surat and predicting whether they require a website.

The project involved collecting and preprocessing business data, performing exploratory data analysis, developing interactive Power BI dashboards, and implementing multiple machine learning models for prediction. Among the evaluated models, the Decision Tree algorithm achieved the highest accuracy of 92.81%, making it the most suitable model for this dataset.

The analysis demonstrated that a large proportion of businesses still operate without a website despite maintaining high customer ratings. These findings highlight significant opportunities for digital transformation and support data-driven decision-making for businesses and digital service providers.

Overall, the project demonstrates how data analytics and machine learning can be combined to generate meaningful business insights and assist organizations in identifying businesses with strong digital growth potential.